JÖNKÖPING UNIVERSITY
School of Engineering

TARI29 Artificial Intelligence, 7.5 credits, Atumn 2023

Exam 2

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Date: 7 January 2024 14:00–19:00

Allowed aids:

- VS Code and SWI-Prolog installed at the computers in the exam room
- SWI-Prolog documentation provided as PDF
- The logical equivalence sheet provided as PDF
- Other material provided as PDF

However, you are not allowed to use any other form of assistance. Specifically, it is of course strictly prohibited to collaborate with other students or use online resources during the exam.

Instructions for answering questions:

- Each question will provide a task description and the maximum amount of points that can be awarded for a correct answer.
- Type in your answers directly in the Inspera system.
- If a question mentions that you can submit a drawing or other type of writing on paper, you can get paper sheets from the personnel in the exam room. Then you need to hand in your sheets to the personnel. The paper sheets will be delivered to the examiners after the exam. The paper sheets will be delivered to the examiners after the exam. If you write on paper, **write in legible (clear) handwriting** so that your answer is well readable.
- For Prolog problems, write and debug your code in the VS Code environment. Then copy the Prolog code of your solution from VS Code and paste it directly in the text field of the corresponding question.

Grading:
The maximum total score for the written examination is 70 points, and the grading is:

3	28–41.5	points
4	42–55.5	points
5	56–70	points

The distribution of points among the exam sections:

Section:	Search	Game Theory	MDP	Logic	Prolog	Total
Points:	12	12	9	14	23	70

A Search (12 points)

1. (4 points) **Search: General Problem Solver–Task 1**.

Looking at toy problems like water jugs etc., where you want to find the shortest solution, is the following statement true or false? Give a very brief motivation for your answer.

- The first solution found by breadth first is always an optimal solution.

Solution and Marking:

Here, the motivation is the key part to score the points. On this question, you can argue for both answers. Please note, however, that it says "toy problems like water jugs..." so the question is not about graph search in general, but rather problems where you want to find the shortest possible solution in number of moves; or put another way, problems where every move costs 1.

The statement is true: BF does find the shortest solution in number of moves.

Marking:

- 1 point if very little motivation is given that does not allow making a good assessment of the answer
- 2 points if the answer is incomplete, e.g. not specified what is meant by the optimal/fastest solution in the answer.
- 4 points for a complete answer with good motivation

2. (4 points) **Search: General problem solver–Task 2**.

Looking at toy problems like water jugs etc., where you want to find the shortest solution, is the following statement true or false? Give a very brief motivation for your answer.

- The first solution found by iterative deepening search is always an optimal solution.

Solution and Marking:

Here, the motivation is the key part to score the points. On this question, you can argue for both answers. Please note, however, that it says "toy problems like water jugs..." so the question is not about graph search in general, but rather problems where you want to find the shortest possible solution in number of moves; or put another way, problems where every move costs 1.

The statement is true: IDS will work one level at the time, so the first solution will contain the least possible number of moves.

Marking:

- 1 point if very little motivation is given that does not allow making a good assessment of the answer
- 2 points if the answer is incomplete, e.g. not specified what is meant by the optimal/fastest solution in the answer.
- 4 points for a complete answer with good motivation

3. (4 points) **Search: General problem solver–Task 3**.

Looking at toy problems like water jugs etc., where you want to find the shortest solution, is the following statement true or false? Give a very brief motivation for your answer.

- In the general problem solver framework, you never have to add a state which is already in OPEN or CLOSED to OPEN, even if you reach it from another node.

Solution and Marking:

Here, the motivation is the key part to score the points. On this question, you can argue for both answers. Please note, however, that it says "toy problems like water jugs..." so the question is not about graph search in general, but rather problems where you want to find the shortest possible solution in number of moves; or put another way, problems where every move costs 1.

The statement is true: GPS works by only adding nodes that are not in either OPEN or CLOSED.

Marking:

- 0 point for an incorrect answer with no motivation
- 1 point for an incorrect answer with some motivation
- 2 points for a correct answer, either with very little or no motivation that does not allow making a good assessment of the answer or with motivation that is irrelevant for the question
- 4 points for a complete answer with good motivation

Marking:

B Game Theory (12 points)

4. (6 points) **Game Theory: Task 1**

Consider the payoff matrix below for a game in normal form.

		Player B		
		X	**Y**	**Z**
	X	**5,5**	**3,0**	**0,2**
Player A	**Y**	**4,1**	**2,1**	**1,0**
	Z	**0,0**	**2,4**	**4,5**

Are there any pure Nash Equilibria or Pareto Optimal outcomes in the game? If so, what are they (and why)?

Solution and Marking:

In a Nash Equilibrium, no player wants to change to another possible action, since the payoff will be smaller.

(**X=5**,**X=5**) is a Nash Equilibrium since **Player A** doesn't want to choose action **Y=4** or **Z=0**, since action **X=5** yields a higher payoff, and **Player B** doesn't want to choose action **Y=0** or **Z=2**, since action **X=5** yields a higher payoff.

The same applies to (**Z=4**,**Z=5**). **Player A** doesn't want to choose action **X=0** or **Y=1**, since action **Z=4** yields a higher payoff, and **Player B** doesn't want to choose action **X=0** or **Y=4**, since action **Z=5** yields a higher payoff. Therefore, (**Z=4**,**Z=5**) is a Nash Equilibrium, as well.

Pareto optimality is defined as:

- If an outcome o is at least as good for each agent as another outcome $o^{'}$, where at least one agent strictly prefers o over $o^{'}$, then o Pareto dominates $o^{'}$.
- An outcome $o*$ is Pareto optimal if there isn't any other outcome that Pareto dominates $o*$.

Both (**X=5**,**X=5**) and (**Z=4**,**Z=5**) dominate the other outcomes because at least one agent strictly prefers one of those outcomes compared to any of the rest. However, only (**X=5**,**X=5**) is a Pareto Optimal outcome, since it Pareto dominates (**Z=4**,**Z=5**). In that case, **Player A** strictly prefers **X=5** over **Z=4**.

Marking:

- -1p for each incorrect or missing Nash Equilibria. ($\times 2$)
- -1p for insufficient justification of Nash Equilibria. ($\times 2$)
- -1p for each incorrect or missing Pareto Optimal outcome.
- -1p for insufficient justification of Pareto Optimal outcomes.

5. (2 points) **Game Theory: Task 2**

Let us assume the following scenario. Volvo Group and Scania, two leading Swedish automobile manufacturers, produce new cars with similar features. They can produce high or low quantities. Based on their decision, the following cases may occur:

- if they cooperate and agree to produce at low levels, then limited supply will translate into a high price for the product on the market and substantial profits of 90 million SEK for both companies
- if they defect and produce at high levels, the market will be swamped and result in a low price for the product and consequently lower profits for both, equal to 30 million SEK
- but if one cooperates (i.e. produces at low levels) and the other defects (i.e. surreptitiously produces at high levels), then the former just break even (i.e., no profits) while the latter earns a profit of 120 million SEK.

Represent the above situation as a game in its Normal Form. What is the dominant strategy of each company and why? Can we predict what would be the outcome of this game? Do we make any assumptions to make this prediction?

Solution and Marking:

We represent the game in its Normal Form as shown in the payoff matrix below:

		Scania	
		Cooperate	**Defect**
Volvo Group	**Cooperate**	**9**,**9**	**0**,**12**
	Defect	**12**,**0**	**3**,**3**

About dominance, we know the following:

- A **dominant strategy** is a strategy that **dominates all other strategies**.
- A strategy profile that consists of **dominant strategies for each player**, must necessarily constitute a **Nash Equilibrium** (no player wants to change strategy). Two types of dominance exist:
  - **Strict dominance:** For every strategy that the other players can play, if the utility of strategy s_i for player i is larger than any other strategy s'_i for player i, then there is a strict dominance of strategy s_i over strategy s'_i.
  - **Weak dominance:** The same as the above, but strategy s_i is larger or equal to strategy s'_i, which means that there is a weak dominance of strategy s_i over strategy s'_i.
- An **equilibrium that only consists of strict dominant strategies** must be **unique**.

For **Volvo Group**, strategy **Cooperate** is strictly dominated by strategy **Defect**, since:

- if $a_{Scania} = Cooperate$ then $a^*_{VolvoGroup} = Defect$, where $u_{VolvoGroup}(Defect, Cooperate) = 12 > u_{VolvoGroup}(Cooperate, Cooperate) = 9$.
- if $a_{Scania} = Defect$ then $a^*_{VolvoGroup} = Defect$, where $u_{VolvoGroup}(Defect, Defect) = 3 > u_{VolvoGroup}(Cooperate, Defect) = 0$.

For **Scania**, strategy **Cooperate** is strictly dominated by strategy **Defect**, since:

- if $a_{VolvoGroup} = Cooperate$ then $a^*_{Scania} = Defect$, where $u_{Scania}(Cooperate, Defect) = 12 > u_{Scania}(Cooperate, Cooperate) = 9$.
- if $a_{VolvoGroup} = Defect$ then $a^*_{Scania} = Defect$, where $u_{Scania}(Defect, Defect) = 3 > u_{Scania}(Defect, Cooperate) = 0$.

Therefore, the dominant strategy for **Volvo Group** is **Defect**, and for **Scania** is **Defect**, as well.

Considering that for an outcome that constitutes a **Nash Equilibrium**, **no agent can increase its utility** by changing to another action, we predict that the game's outcome will be (**Defect**,**Defect**).

To predict that outcome, we assumed that **both companies** would act **rationally**.

Marking:

- -0.5p for incorrect or missing Normal Form.
- -0.5p for each incorrect or missing dominant strategy.
- -0.5p for the incorrect or missing outcome of the game.
- -0.5p for incorrect or missing assumptions.

6. (4 points) **Game Theory: Task 3**

Calculate a Nash Equilibrium with mixed strategies from the payoff matrix below and state the value of the game at this equilibrium. Use the method of assuming the opponent should be indifferent between its actions. Show all your calculations.

		Player B	
		X	**Y**
Player A	**X**	**1**,**1**	**3**,**0**
	Y	**4**,**1**	**2**,**2**

Solution and Marking:

We can denote the probability that **Player A** plays strategy **X** with **p** (and **1-p** for strategy **Y**) and that **Player B** plays strategy **X** with probability **q** (and **1-q** for strategy **Y**).

			Player B	
			q	**1-q**
			X	**Y**
Player A	**p**	**X**	**1**,**1**	**3**,**0**
	1-p	**Y**	**4**,**1**	**2**,**2**

We then reason that **Player B** has chosen his or her probabilities **q** and **1-q** in such a way that **Player A** is indifferent between strategy **X** and **Y**. Likewise, we reason that **Player A** has chosen his or her probabilities **p** and **1-p** in such a way that **Player B** is indifferent between strategy **X** and **Y**. This gives us the equations below:

Player A (row X = row Y)

$$\begin{aligned} 1q + 3(1-q) &= 4q + 2(1-q) \\ q + 3 - 3q &= 4q + 2 - 2q \\ q - 3q - 4q + 2q &= -3 + 2 \\ -4q &= -1 \\ q &= 1/4 \\ 1 - q &= 3/4 \end{aligned}$$

Player B (column X = column Y)

$$\begin{aligned} 1p + 1(1-p) &= 0p + 2(1-p) \\ p + 1 - p &= 2 - 2p \\ p - p + 2p &= 2 - 1 \\ 2p &= 1 \\ p &= 1/2 \\ 1 - p &= 1/2 \end{aligned}$$

This gives us q = 1/4, 1-q = 3/4 and p = 1/2, 1-p = 1/2. Hence, we have a Nash Equilibrium with the mixed strategies:

- Player A plays strategy X 1/2 of the time and strategy Y 1/2 of the time.
- Player B plays strategy X 1/4 of the time and strategy Y 3/4 of the time.

The value of the game at this mixed-strategy Nash Equilibrium is obtained by multiplying each pay-off for each player in each outcome (cell) with the corresponding probability for Player A and Player B in that outcome, and finally summing the result in each cell (i.e. the expected value):

Player A

$$\begin{aligned} U_A &= 1pq + 3p(1-q) + 4(1-p)q + 2(1-p)(1-q) \\ &= 1pq + 3p - 3pq + 4q - 4pq + 2 - 2q - 2p + 2pq \\ &= -4pq + p + 2q + 2 \\ &= -4 \times 1/2 \times 1/4 + 1/2 + 2 \times 1/4 + 2 \\ &= -1/2 + 1/2 + 1/2 + 2 \\ &= 1/2 + 4/2 \\ &= 5/2 \\ &\approx 2.50 \end{aligned}$$

Player B

$$\begin{aligned}
U_B &= 1pq + 1p(1-q) + 0(1-p)q + 2(1-p)(1-q) \\
&= pq + p - pq + 0q - 0pq + 2 - 2q - 2p + 2pq \\
&= 2pq - p - 2q + 2 \\
&= 2 \times 1/2 \times 1/4 - 1/2 - 2 \times 1/4 + 2 \\
&= 1/4 - 1/2 - 1/2 + 2 \\
&= 1/4 - 1 + 2 \\
&= 1/4 - 4/4 \\
&= 3/4 \\
&\approx 0.75
\end{aligned}$$

The value of the game is, hence: $(U_A, U_B) = (5/2, 3/4) \approx (2.50, 0.75)$.

Marking:

- -1p for an incorrect mixed strategy per player. (Mixed strategies: 2p)
- -1p for an incorrect game value per player. (Value of the game: 2p)

C MDP / Reinforcement Learning (9 points)

7. (3 points) **MDP / Reinforcement Learning: Task 1**

Assume an MDP consists of the two states S_1 and S_2, the two actions A_1 and A_2, a discount factor $\gamma = 1/4$, a reward function R and a transition function T as shown below:

$$R(s, a, s') = \begin{cases} 1/2 & \text{if } s' = S_1 \\ -1/2 & \text{if } s' = S_2 \end{cases}$$

s	a	s'	T(s,a,s')
S_1	A_1	S_1	3/4
S_1	A_1	S_2	1/4
S_1	A_2	S_1	1/2
S_1	A_2	S_2	1/2
S_2	A_1	S_1	1/3
S_2	A_1	S_2	2/3
S_2	A_2	S_1	1/2
S_2	A_2	S_2	1/2

Perform one iteration of value iteration and enter the value for $V_1(S_1)$ in the table below (you don't have to enter a value into the gray cell). Use the initial values for $V_0(s)$ below instead of initializing all values to 0. Show all your calculations.

$$V_{k+1}(s) \leftarrow \max_a \sum_{s'} T(s, a, s') \left[R(s, a, s') + \gamma V_k(s') \right]$$

s	$V_0(s)$	$V_1(s)$
S_1	4	?
S_2	2	

Solution and Marking:

By substituting values into the value iteration update equation we get:

$$V_{k+1}(s) \leftarrow \max_a \sum_{s'} T(s,a,s') \left[R(s,a,s') + \gamma V_k(s')\right]$$

$$V_1(S_1) \leftarrow max \begin{pmatrix} A_1 : T(S_1,A_1,S_1)\left[R(S_1,A_1,S_1) + \gamma V_0(S_1)\right] + T(S_1,A_1,S_2)\left[R(S_1,A_1,S_2) + \gamma V_0(S_2)\right] \\ A_2 : T(S_1,A_2,S_1)\left[R(S_1,A_2,S_1) + \gamma V_0(S_1)\right] + T(S_1,A_2,S_2)\left[R(S_1,A_2,S_2) + \gamma V_0(S_2)\right] \end{pmatrix}$$

$$V_1(S_1) \leftarrow max \begin{pmatrix} A_1 : \frac{3}{4} * \left[\frac{1}{2} + \frac{1}{4} * 4\right] + \frac{1}{4} * \left[-\frac{1}{2} + \frac{1}{4} * 2\right] = \frac{3}{4} * (\frac{1}{2} + 1) + \frac{1}{4} * (-\frac{1}{2} + \frac{1}{2}) = \frac{3}{4} * \frac{3}{2} + 0 = \frac{9}{8} = 1.125 \\ A_2 : \frac{1}{2} * \left[\frac{1}{2} + \frac{1}{4} * 4\right] + \frac{1}{2} * \left[-\frac{1}{2} + \frac{1}{4} * 2\right] = \frac{1}{2} * (\frac{1}{2} + 1) + \frac{1}{2} * (-\frac{1}{2} + \frac{1}{2}) = \frac{1}{2} * \frac{3}{2} + 0 = \frac{3}{4} = 0.75 \end{pmatrix}$$

s	$V_0(s)$	$V_1(s)$
S_1	4	1.125
S_2	2	

Marking:

- -2.5p for mistakes in the formula. (2p+0.5p)
  - -0.125p for each incorrect or missing parameter in the equation. ($16 \times 0.125 = 2$)
  - -0.5p for **any** calculation errors.
- -0.5p for the wrong result (of $V_1(s)$).

8. (6 points) **MDP / Reinforcement Learning: Task 2**

Assume an MDP consists of the two states S_1 and S_2, the two actions A_1 and A_2, and a transition function T as shown below:

s	a	s'	T(s,a,s')
S_1	A_1	S_1	1/4
S_1	A_1	S_2	3/4
S_1	A_2	S_1	1/2
S_1	A_2	S_2	1/2
S_2	A_1	S_1	1/4
S_2	A_1	S_2	3/4
S_2	A_2	S_1	1/2
S_2	A_2	S_2	1/2

Assume Q-learning with a learning rate $\alpha = 1/4$ and without discount (i.e., $\gamma = 1$) is used, and the following episode is observed (i.e. first the agent executes the action A_1 in state

S_1, ending up in state S_1 with the reward *-1*, followed by the agent executing the action A_1 in state S_1, ending up in state S_2 with the reward *1*):

s_1	a_1	r_1	s_2	a_2	r_2	s_3
S_1	A_1	-1	S_1	A_1	1	S_2

Use the initial Q-values Q_0 in the table below, and enter the values for Q_1 and Q_2. Show all your calculations.

s	a	$Q_0(s,a)$	$Q_1(s,a)$	$Q_2(s,a)$
S_1	A_1	1/2	?	?
S_1	A_2	1/8	?	?
S_2	A_1	5/4	?	?
S_2	A_2	1/4	?	?

$$Q_{k+1}(s,a) \leftarrow Q_k(s,a) + \alpha\left[r + \gamma \max_{a'} Q_k(s',a') - Q_k(s,a)\right]$$

Solution and Marking:

By substituting values into the q-value update equation we get:

$$Q_{k+1}(s,a) \leftarrow Q_k(s,a) + \alpha\left[r + \gamma \max_{a'} Q_k(s',a') - Q_k(s,a)\right]$$

First step:

$$\begin{aligned} Q_1(S_1,A_1) &\leftarrow Q_0(S_1,A_1) + \frac{1}{4}\left[-1 + 1\cdot\max_{a'} Q_0(S_1,a') - Q_0(S_1,A_1)\right] \\ &\leftarrow \frac{1}{2} + \frac{1}{4}\left[-1 + 1\cdot\max(\frac{1}{2},\frac{1}{8}) - (\frac{1}{2})\right] = \frac{1}{2} + \frac{1}{4}\left[-1 + \frac{1}{2} - \frac{1}{2}\right] = \frac{1}{2} - \frac{1}{4} = \frac{1}{4} \end{aligned}$$

$Q_1(S_1,A_2) \leftarrow Q_0(S_1,A_2)$ since A_2 was not taken in S_1 during step 1 in the episode
$Q_1(S_2,A_1) \leftarrow Q_0(S_2,A_1)$ since A_1 was not taken in S_2 during step 1 in the episode
$Q_1(S_2,A_2) \leftarrow Q_0(S_2,A_2)$ since A_2 was not taken in S_2 during step 1 in the episode

Second step:

$$\begin{aligned} Q_2(S_1,A_1) &\leftarrow Q_1(S_1,A_1) + \frac{1}{4}\left[1 + 1\cdot\max_{a'} Q_1(S_2,a') - Q_1(S_1,A_1)\right] \\ &\leftarrow \frac{1}{4} + \frac{1}{4}\left[1 + 1\cdot\max(\frac{5}{4},\frac{1}{4}) - \frac{1}{4}\right] = \frac{1}{4} + \frac{1}{4}\left[1 + \frac{5}{4} - \frac{1}{4}\right] \\ &= \frac{1}{4} + \frac{1}{4}\left[1 + \frac{4}{4}\right] = \frac{1}{4} + \frac{1}{4}\cdot 2 = \frac{1}{4} + \frac{2}{4} = \frac{3}{4} \end{aligned}$$

$Q_2(S_1,A_2) \leftarrow Q_1(S_1,A_2)$ since A_2 was not taken in S_1 during step 2 in the episode
$Q_2(S_2,A_1) \leftarrow Q_1(S_2,A_1)$ since A_1 was not taken in S_2 during step 2 in the episode
$Q_2(S_2,A_2) \leftarrow Q_1(S_2,A_2)$ since A_2 was not taken in S_2 during step 2 in the episode

s	a	$Q_0(s,a)$	$Q_1(s,a)$	$Q_2(s,a)$
S_1	A_1	1/2	1/4	3/4
S_1	A_2	1/8	1/8	1/8
S_2	A_1	5/4	5/4	5/4
S_2	A_2	1/4	1/4	1/4

Marking:

- -2p for mistakes in the formula of the first step. (1.4p+0.6p)
  - -0.2p for each incorrect or missing parameter in the equation. ($7 \times 0.2 = 1.4$)
  - -0.6p for **any** calculation errors.
- -2p for mistakes in the formula of the second step. (1.4p+0.6p)
  - -0.2p for each incorrect or missing parameter in the equation. ($7 \times 0.2 = 1.4$)
  - -0.6p for **any** calculation errors.
- -1p for calculating anything apart from $Q_1(S_1, A_1)$ in the first step.
- -1p for calculating anything apart from $Q_2(S_1, A_1)$ in the second step.

D Logic (14 points)

9. (4 points) **Propositional Logic: Wumpus world—Task 1**

Consider a Wumpus world, where the agent has visited the following squares and has the following perceptions:

[1, 1]: Smell, but no Breeze

[1, 2]: no Smell or Breeze

[1, 3]: Breeze, but no Smell

The agent is currently at square [2,2] and perceives both a Smell and a Breeze.

How can the agent infer that the Wumpus is at square [2, 1]? Describe the reasoning in a mixture of words and formulas. You may need to take into account some general facts about the Wumpus world, as described in Russell & Norvig.

Solution and Marking:

Answers containing reasoning which refer to formulas are OK.

Marking:

- 0 point if no reasoning or no answer is given
- 1 point if the answer only mentions general facts about the wumpus world

- 2 points if the world rule about squares next to a Wumpus being smelly is stated, with the inference that the Wumpus is in either [1,2] or [2,1]
- 2 points if the fact that square [1,2] is ruled out because it was visited and the agent did not die is mentioned somehow.
- -2 points for an incomplete reasoning

10. (5 points) **Logic: Propositional logic—Task 2**

Consider a Wumpus world, where the agent has visited the following squares and has the following perceptions:

[1, 1]: Smell, but no Breeze

[1, 2]: no Smell or Breeze

[1, 3]: Breeze, but no Smell

The agent is currently at square [2,2] and perceives both a Smell and a Breeze.

If the inference in the previous question (Propositional logic—Task 1) were to be implemented in propositional logic and proved using truth tables, how many rows would that truth table have?

Hint: Think carefully about which facts and general rules of the world are needed! Give a thorough motivation/explanation for your answer. Note that you do not have to construct the actual truth table, but only state which propositions need to be included and how big the truth table then becomes.

Solution and Marking:

At least the propositional symbols for the rule for stench in square [1,1] need to be used. So, the minimum is 3 symbols:

$S_{1,1} \leftrightarrow (W_{1,2} \vee W_{2,1})$

This gives a truth table of 8 rows.

Marking:

- 0 points if no answer is given or for an (incorrect) truth table with NO explanation
- 0 points for answers saying that only two symbols, i.e. 4 rows, are needed, since this ignores the conclusion, which has to be included in the table for the proof.
- 1 point for stating how big the truth table becomes
- 2 points for propositional symbol for given facts about each square
- 2 points for propositional symbol for general facts about the Wumpus world

11. (5 points) **Logic: Propositional logic—Task 3**

Consider a Wumpus world, where the agent has visited the following squares and has the following perceptions:

[1, 1]: Smell, but no Breeze

[1, 2]: no Smell or Breeze

[1, 3]: Breeze, but no Smell

The agent is currently at square [2,2] and perceives both a Smell and a Breeze.

Can the agent infer that square [3, 2] is safe, i.e. contains neither a Wumpus nor a pit? Either carry out the inference, using only propositional logic, or give a thorough explanation of why the agent cannot make the inference.

Solution and Marking:

Safety cannot be inferred, since there might be a pit in [3,2].

Marking:

- 0 points if no clear answer or wrong answer is given
- 0 points if incorrect explanation or just random statements are given
- 0 points for a correct answer but with irrelevant explanation
- 3.5 points if inference uses general facts about the wumpus world, but the inference mentions incorrect squares
- 5 points if the answer states that safety cannot be inferred, since there might be a pit in [3,2].

E Prolog (23 points)

12. (8 points) **Prolog: Constraint Logic Programming (CLP)**

Max has 25 kr for buying candies. Gumballs cost 3 kr, snickers cost 10 kr, toffees cost 9 kr and liquorice costs 7 kr, all prices are per candy. What alternatives does Max have for buying candies? Assume that Max spends the entire sum of 25 kr. Candies can be mixed in any way.

Write a Prolog program that solves the problem using CLP(FD). You need to find all concrete solutions.

Solution and Marking:

```
:- use_module(library(clpfd)).
candies_sol2([G, S, T, L]) :-
    % Gumballs, snickers, toffees, liquorice
    [G, S, T, L] ins 0..25,
    G*3 + S*10 + T*9 + L*7 #= 25,
    label([G, S, T, L]).
```

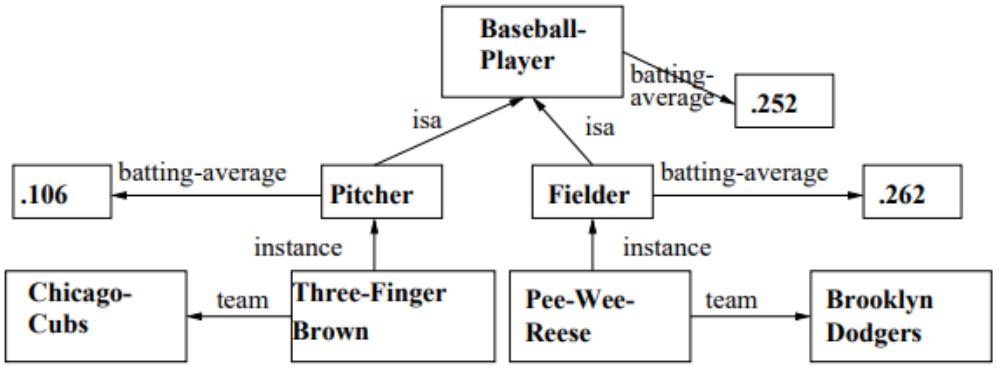
Marking:

- 2 points if variables are correctly declared/used

- −1 point if there are too many variables
- 0 points if variables are declared/used incorrectly or no variable at all
- 2 points, if the domains of variables are defined and in a correct way
- −1 point for incorrect domains of variables
- 2 points if an arithmetic expression is correct
- 0 point for an incorrect expression
- 2 points if all conrete solutions are found correct
- −2 points if `all_different(Vars)` is used—the variables can be the same here (in contrast to a cryptoarithmetic puzzle)

13. (7 points) **Prolog: Semantic Networks**

In this task you will use semantic network to encode several facts about baseball. The following diagram shows an example of a simple semantic network (source: https://www.professional-ai.com/difference-between-semantic-net-and-frame.html).



The task is to represent this semantic network with Prolog clauses.

Hint: Each triple 'box–arrow–box' from the diagram will become a Prolog clause with a binary predicate corresponding to the label of the arrow between the boxes. The boxes themselves will become the arguments.

Solution and Marking:

```
isa(pitcher, baseball_player).
isa(fielder, baseball_player).
instance(three_finger_brown, pitcher).
instance(pee_wee_reese, fielder).
team(three_finger_brown, chicago_cubs).
team(pee_wee_reese, brooklyn_dodgers).
batting_average(pitcher, 0.106).
```

```
batting_average(fielder, 0.262).
batting_average(baseball_player, 0.252).
```

Marking:

- 1.5 point for each of the following
  - the two isa-clauses
  - the two instance-clauses
  - the two team-clauses
- 2.5 points for the three batting_average-clauses
- 0 points for an incorrect solution

14. (4 points) **Prolog: Data structures—Task 1**

Define the predicates `binarytree(Object)` to recognize whether `Object` is a binary tree written in the notation introduced in the lecture as well as in the textbook.

The notation is as follows:

- Let the atom `nil` represent the empty tree.
- Let the functor be `t` so the tree that has a root `X`, a left subtree `L` and a right subtree `R` is represented by the term `t(L, X, R)`.

Use this example from the lecture to test your solution:
`t( t(nil, b, nil), a, t( t(nil, d, nil), c, nil ) )`

Solution and Marking:

```
binarytree(nil).
binarytree( t(Left, Root, Right) ) :-
    binarytree(Left),
    binarytree(Right).
```

Marking:

- 1 point for each of the following
  - the recursion terminating case with `nil`
  - a recursive rule head with functor `t`
  - each subtree is visited
- −3 points if a list of nodes is constructed

15. (4 points) **Prolog: Data structures—Task 2**

Define the procedure `height(BinaryTree, Height)` to compute the height of a binary tree. Assume that the height of the empty tree is 0, and that of a one-element tree is 1.

The notation is as follows:

- Let the atom `nil` represent the empty tree.
- Let the functor be `t` so the tree that has a root `X`, a left subtree `L` and a right subtree `R` is represented by the term `t(L, X, R)`.

Use this example from the lecture to test your solution:
`t( t(nil, b, nil), a, t( t(nil, d, nil), c, nil ) )`

Solution and Marking:

```
% Note: max can be different
max(X, Y, X):- X >= Y.
max(X, Y, Y):- X < Y.

height(nil, 0).
height( t(Left, Root, Right), H) :-
    height(Left, LH),
    height(Right, RH),
    max(LH, RH, MH),
    H is 1 + MH.
```

Marking:

- 1 point for each of the following
  - the recursion terminating case with `nil`
  - a recursive rule head with functor `t`
  - each subtree is visited
  - `max` is calculated
- −1 point if unneeded clauses aresued
- −1 point if non-Prolog arithemtic is used
- −2 points if predicate names are variables names